

A REPORT OF 4-6 MONTHS INDUSTRIAL TRAINING

at

EXCELLENCE TECHNOLOGY

**SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR**

THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

(Electronics and Computer Engineering)



JAN-JUNE 2026

SUBMITTED BY:

PARAS BANSAL

12205022

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

PUNJABI UNIVERSITY, PATIALA

CERTIFICATE



Ref.No.: ET/2090

Date: 20 Apr 2026

To Whom It May Concern

Subject: Training Pursuing Certificate

This is to certify that Mr. Paras Bansal S/o Mr. Anil Kumar of Punjabi University Patiala is currently undergoing training in the field of Cyber Security at Excellence Technology.

The training commenced on 02 Feb 2026 and is ongoing as of the date of 31 July 2026. During this period, the trainee has been actively involved in learning and working on various concepts and practical applications related to Cyber Security.

This certificate is being issued upon the request of the candidate as proof of their ongoing training with our organization.

We wish them all the best for their future endeavors.

For Excellence Technology

Mr. Deepak Kashyap
Manager



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E : info@excellencetechnology.in, W : www.excellencetechnology.in
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CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the B. Tech Industrial Project Report entitled “**SOC SIMULATION LAB CREATED USING VIRTUAL MACHINES AND SIEM TOOLS**”, in partial fulfilment of the requirements for the award of the Bachelor of Technology in Electronics & Computer Engineering and submitted to the Department of Electronics & Communication Engineering of Punjabi University, Patiala is an authentic record of our work carried out during a period from 02 FEB 2026 to 31 JULY 2026.

Paras Bansal

Signature of the Student

The final presentation and viva-voce of _____
_____ has been held on
_____ and accepted.

Signature of the Examiner

ACKNOWLEDGEMENT

I would like to express my sincere appreciation and gratitude to all those who contributed to the successful completion of this project titled “**SOC Simulation Lab Created Using Virtual Machines and SIEM Tools.**”

I would like to extend my heartfelt gratitude to my Training Coordinator, **Dr. Beant Kaur, Assistant Professor, Department of Electronics and Communication Engineering, Punjabi University, Patiala**, for her unwavering support, guidance, and constant availability throughout the course of this training. Her continuous assistance in addressing queries, providing project-related insights, and conveying important updates played a crucial role in the successful completion of this work

I am especially thankful to my respected project guide for their valuable guidance, constant encouragement, and insightful suggestions throughout the development of this project. Their support has been instrumental in enhancing my understanding of cybersecurity concepts and the practical implementation of SIEM-based monitoring systems.

I would also like to extend my gratitude to the faculty members of my institution for providing the necessary academic support and resources required to complete this project successfully.

Furthermore, I am thankful to my peers for their cooperation and constructive discussions, which helped me overcome various challenges during the project.

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PART -1 INDUSTRIAL TRAINING REPORT

EXCELLENCE TECHNOLOGY

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CHAPTER 1

1.1 INTRODUCTION TO ORGANISATION



Figure 1.1 Company logo

EXCELLENCE TECHNOLOGY is one of the emerging and reliable companies, well-established to deliver customer-centric B2B and B2C services along with innovative and indigenous solutions across multiple industry domains. Our expertise spans Digital Services, IT Solutions, Business Consulting, and Technology-driven services. With a team of skilled professionals, we are committed to delivering high-quality, efficient, and result-oriented solutions that meet the evolving needs of our clients.

Established with a vision to empower businesses, **EXCELLENCE TECHNOLOGY** aims to help organizations grow, transform, and establish a strong presence in the digital world. Over the years, we have consistently focused on enhancing customer experience through advanced technologies, strategic development, and a dedication to excellence. Today, we stand among trusted service providers in the IT and Digital Services sector, known for delivering reliable and scalable solutions tailored to client requirements.

- We focus on **understanding client requirements** to deliver customized and effective business solutions.
- Our **cost-effective approach** ensures value-driven services without compromising quality.
- We maintain **complete confidentiality and data security** in every project we handle.
- Our **end-to-end service model** ensures seamless execution and long-term client satisfaction.

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1.2 COMPANY OUTCOME

Excellence Technology aims to deliver impactful and result-oriented solutions that contribute to the growth and success of its clients. The company focuses on building long-term relationships by ensuring quality service, innovation, and customer satisfaction.

The key outcomes of the organization include the development of efficient digital systems, enhanced cybersecurity practices, and improved business performance for clients. The company also contributes to the development of skilled professionals by providing practical training and exposure to real-world technologies.

1.2.1 Vision

The vision of Excellence Technology is to become a leading provider of innovative and technology-driven solutions by delivering high-quality services that meet the evolving needs of businesses. The company aims to establish a strong presence in the IT industry by focusing on excellence, reliability, and continuous improvement.

1.2.2 Mission

The mission of Excellence Technology is to provide cost-effective, secure, and efficient solutions that help organizations achieve their business goals. The company is committed to maintaining high standards of quality, ensuring data security, and delivering customer-centric services. It also aims to empower individuals by providing practical training and enhancing technical skills required in the modern IT industry.

1.3 TECHNOLOGIES LEARNED DURING TRAINING

1.3.1 VMware

VMware, now a Broadcom company, is a leading provider of virtualization and cloud computing software, enabling businesses to run multiple virtual machines (VMs) on a single physical server.

VMware Workstation Installation on Windows:

1. Download VMware Workstation

- Go to the official VMware site and download **VMware Workstation Pro** or **VMware Workstation Player** (Player is free for personal use).
- Choose the Windows installer (.exe file).

2. Run the Installer

- Double-click the downloaded file.
- Accept the license agreement.
- Select installation options:
 - **Enhanced Keyboard Driver** (recommended).
 - **Check for product updates** (optional).
- Choose installation folder (default is fine).

3. Complete Installation

- Click **Install** and wait for setup to finish.
- Restart your computer if prompted.

4. Initial Setup

- Launch VMware Workstation.
- Enter license key (if using Pro).
- For Player, you can skip this step.

5. Create a Virtual Machine

- Click **Create a New Virtual Machine**.
- Choose **Typical (recommended)**.
- Select installation source:
 - ISO file (e.g., Windows/Linux installer).
 - Physical disc.
- Assign VM name and location.

6. Configure Hardware

- Allocate CPU cores and RAM (e.g., 2 cores, 4 GB RAM for light use).
- Set disk size (20–40 GB recommended).
- Choose **Store virtual disk as a single file** for better performance.

7. Install Guest OS

- Power on the VM.
- Follow the OS installation wizard (Windows/Linux/etc.).
- Install VMware Tools inside the guest OS for better performance (drivers, clipboard sharing, drag-and-drop).

1.3.2 OPERATING SYSTEMS USED

- **Kali Linux 2025.4 (VMware AMD64)** A penetration testing–focused Linux distribution widely used by security professionals. It comes preloaded with tools for ethical hacking, digital forensics, and vulnerability assessment. Optimized for VMware, it supports 64-bit architecture for smooth virtualization.
- **Metasploitable2 Linux** A deliberately vulnerable Linux virtual machine designed for practicing penetration testing. It provides a safe environment to test exploits, security tools, and vulnerability scanning techniques. Commonly used in cybersecurity labs and training.
- **Windows 11 x64** A widely used desktop operating system providing stability, compatibility, and user-friendly features. Supports 64-bit architecture for better performance and multitasking. Commonly used in both personal and professional environments.

1.3.3 USEFUL LINUX COMMANDS

File and Directory Management

- **ls -la**: Lists all files, including hidden, in long format.
- **pwd**: Shows current directory path.
- **cd <dir>**: Changes directory.
- **mkdir <dir>**: Creates a new directory.
- **rmdir <dir>**: Remove an empty directory.
- **touch <file>**: Creates an empty file.
- **cp <src> <dest>**: Copies files/directories.
- **mv <src> <dest>**: Moves or renames files/directories.
- **rm -r <file>**: Deletes files/directories.
- **cat <file>**: Displays file contents.
- **less <file>**: View file content one page at a time (use 'q' to quit).
- **nano / vim / gedit <file>**: Edit files in terminal.
- **find . -name <file>**: Searches for files.
- **grep <pattern> <file>**: Searches for text within files.
- **mount /dev/sdb1 /mnt**: Mount a filesystem.
- **unmount /mnt**: Unmount a filesystem.

System Information and Monitoring

- **top**: Displays real-time system processes.
- **hostnamectl**: system info.
- **lscpu**: CPU architecture info.
- **lsblk**: list disk & partition.
- **ps aux**: Shows all running processes.
- **kill <PID>**: Terminates a process.
- **df -h**: Checks disk space usage.
- **uname -a**: key details about your Linux system.
- **uptime**: Shows system uptime.
- **free -h**: Displays memory usage.
- **journalctl**: View system logs.

Networking

- **ping <host>**: Tests network connectivity.
- **traceroute <hostname or ip>**: Trace the path packets take to a network host.
- **ip a / ifconfig**: Shows network interfaces and IP addresses.
- **ssh <user>@<host>**: Connects to a remote machine.
- **scp <file> <user>@<host>:<path>**: Securely copies files over SSH.
- **wget [http://example.com/file.zip]**: Download files from the server.
- **curl <url>**: retrieve the HTML content of the specified URL and display it in the terminal.

User and Permissions Management

- **whoami**: Shows the current user.
- **id**: Display user identity information.
- **sudo <command>**: Runs commands with root privileges.
- **chmod 755 <file>**: Changes file permissions.
- **chown <user>:<group> <file>**: Changes file ownership.
- **Su <user>**: to login as a user
- **passwd <user>**: Changes user passwords.

1.3.4 LINUX DIRECTORIES

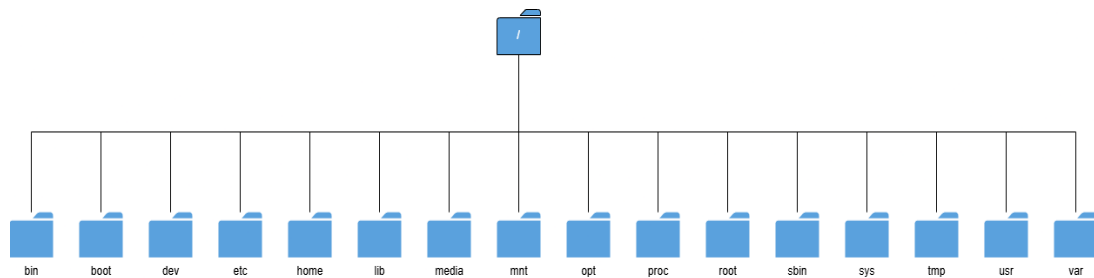


Figure 1.2 Linux Directories

- **/ (Root):** The top-level directory for the entire filesystem.
- **/bin & /usr/bin:** Essential and user command binaries (e.g., ls, cd).
- **/boot:** Bootloader files, kernel, and system startup files.
- **/dev:** Device files representing hardware (e.g., hard drives, terminals).
- **/etc:** System-wide configuration files.
- **/home:** User home directories for storing personal files.
- **/lib & /usr/lib:** Shared libraries required by system binaries.
- **/media & /mnt:** Mount points for removable media (USB, CD-ROM) and temporarily mounted filesystems.
- **/opt:** Optional or third-party software packages.
- **/proc & /sys:** Virtual filesystems providing process and kernel information.
- **/root:** Home directory for the root user (distinct from /).
- **/sbin & /usr/sbin:** System administration binaries (e.g., ifconfig, fdisk).
- **/tmp:** Temporary files created by system/users, often cleared on reboot.
- **/usr:** Contains user-related programs, utilities, and shared resources.
- **/var:** Variable data files, such as system logs (/var/log), mail, and printer queues.

1.3.5 USER AND GROUP

- **Users:** Individual accounts (human or service) that interact with the system, each identified by a unique User ID (UID).
- **Groups:** Collections of users that simplify permission management. Assigning permissions to a group grants the same access to all its members. Each group has a unique Group ID (GID).
- **Types of Users:**
 - **Root (Superuser):** Has full system control and access; used for administrative tasks. Best practices recommend using sudo for elevated privileges rather than logging in directly as root.
 - **Regular Users:** Standard accounts with limited access, designed for daily tasks to maintain system integrity.
 - **System/Service Accounts:** Non-human accounts used by background services (e.g., MySQL, nginx) with minimal privileges.
- **Types of Groups:**
 - **Primary Group:** Every user belongs to one primary group (typically with the same name as the user). New files created by the user are automatically assigned to this group.
 - **Secondary Groups:** Additional groups a user can join to inherit extra permissions.

Essential Commands

Administrative commands typically require sudo privileges.

User Management

- `useradd <username>`: Creates a new user account.
- `passwd <username>`: Sets or changes a user's password.
- `usermod <options> <username>`: Modifies user account properties (e.g., change the primary group with `-g`, add to supplementary groups with `-aG`, change the home directory with `-d`).
- `userdel -r <username>`: Deletes a user account. The `-r` option also removes their home directory and mail spool.
- `id <username>`: Displays the user's UID, GID, and group memberships.

1.3.6 ROOTKIT DETECTION

A **rootkit** is a malicious software that allows an unauthorized user to get access to a system and to its restricted software. These rootkits may contain keyloggers, credential stealers etc.

Key Differences and Usage:

- **Rkhunter (rkhunter -c):** An advanced tool that updates its database to detect both known and unknown rootkits, providing detailed reports on potential vulnerabilities like suspicious shared memory or open ports.
- **Chkrootkit (chkrootkit):** A lighter, faster, and often older tool that focuses on detecting known rootkits, worms, and LKM (Loadable Kernel Module) hacks, requiring manual inspection of results.

Installation & Execution:

- **Install:** `sudo apt install rkhunter chkrootkit`
- **Scan:**
 - `sudo rkhunter --check`
 - `sudo chkrootkit`

Both tools are designed to detect, not remove, infections. They are most effective when run as root, preferably from a trusted live system to avoid reliance on potentially compromised system binaries.

1.3.7 NMAP

Nmap (Network Mapper) is an open-source tool used for network discovery, security auditing, and vulnerability management. It identifies active hosts, open ports, running services (and their versions), and operating systems on a network.

The basic syntax for Nmap is `nmap <options> <target>`. Below are common flags used for different scanning goals:

Fundamental Nmap Commands

- **Basic Scan:** `nmap <target>` - Scans the top 1000 ports on the target.
- **Stealth SYN Scan:** `sudo nmap -sS <target>` - Performs a stealthy scan without completing the TCP handshake.
- **Version Detection:** `nmap -sV <target>` - Identifies service/daemon versions running on ports.
- **OS Detection:** `sudo nmap -O <target>` - Attempts to identify the target's operating system.

- **Aggressive Scan:** `nmap -A <target>` - Enables OS detection, version detection, script scanning, and traceroute.

Port and Host Specification

- **Specific Ports:** `nmap -p 80,443 <target>` - Scans specific ports.
- **Port Range:** `nmap -p 1-65535 <target>` - Scans all TCP ports.
- **Fast Scan:** `nmap -F <target>` - Scans fewer ports than the default scan.
- **Ping Scan:** `nmap -sn <network_address>` - Lists live hosts without scanning ports.
- **Exclude Hosts:** `nmap <network> --exclude <host1,host2>` - Skips specific hosts during a scan.

Targeting Methods

- **Single IP:** `nmap 192.168.1.1.`
- **Subnet:** `nmap 192.168.1.0/24.`
- **IP Range:** `nmap 192.168.1.1-100.`
- **Input File:** `nmap -iL targets.txt` - Scans targets listed in a file.

Common Use Cases

- **Find Vulnerabilities:** `nmap --script vuln <target>` - Runs default vulnerability scripts.
- **UDP Scan:** `sudo nmap -sU <target>` - Scans for open UDP ports.
- **Disable Ping:** `nmap -Pn <target>` - Scans even if the host ignores ICMP requests.

```
(pasta@aastha)-[~]
$ sudo nmap 192.168.1.11 -p80,443,449,138
Starting Nmap 7.94 ( https://nmap.org ) at 2024-02-21 18:54 IST
Nmap scan report for 192.168.1.11 (192.168.1.11)
Host is up (0.000033s latency).

PORT      STATE SERVICE
80/tcp    closed http
138/tcp   closed netbios-dgm
443/tcp    closed https
449/tcp    closed as-servermap

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

Figure 1.3 NMAP

1.3.8 MSFCONSOLE

Msfconsole is the primary, all-in-one command-line interface for the [Metasploit Framework \(MSF\)](#), designed for penetration testing to discover, exploit, and validate vulnerabilities. It enables users to search for exploits, configure payloads, and manage sessions efficiently via tab completion and database integration.

Key Features and Usage:

- **Startup:** Run msfconsole in the terminal; use msfconsole -q for quiet mode (no banner).
- **Database Support:** Use sudo msfdb run to start the PostgreSQL database and msfconsole together.
- **Basic Commands:**
 - `help`: Displays available commands.
 - `search <keyword>`: Finds modules (e.g., `search eternalblue`).
 - `use <module>`: Selects a module to use.
 - `show options`: Lists required settings for the selected module.
 - `set <option> <value>`: Configures a variable (e.g., `set RHOSTS <IP>`).
 - `exploit / run`: Launches the module.
 - `back`: Exits the current module context.

Common Workflows:

- **Scanning:** Use auxiliary modules for reconnaissance.
- **Exploitation:** Select an exploit, set payload (set PAYLOAD), configure target (set RHOSTS), and exploit.
- **Post-Exploitation:** Manage connections via sessions for privilege escalation or data theft.

1.3.9 WIRESHARK

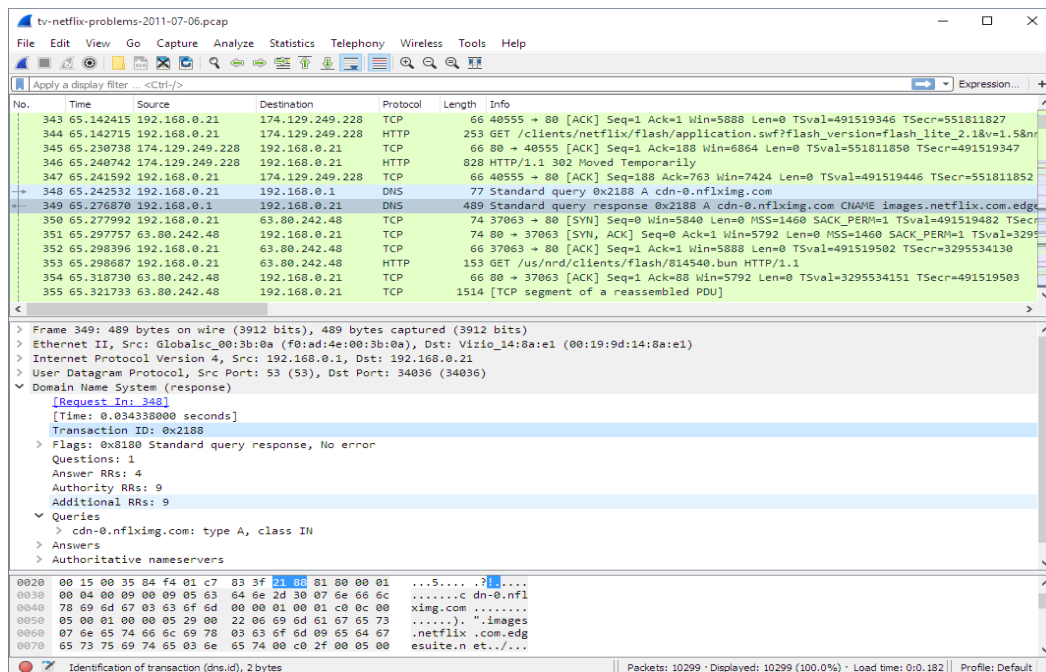


Figure 1.4 WIRESHARK

It is a widely used **open-source network protocol analyzer** that allows users to capture and inspect network traffic in real time. It is commonly used for **network troubleshooting, security analysis, and protocol development**. Wireshark provides deep visibility into packet-level data, helping analysts understand how data flows across a network.

The basic workflow in Wireshark involves selecting a network interface, capturing packets, and applying filters for analysis.

- **Key Features and Functions**
 - **Packet Capture:** Captures live network traffic from interfaces like Ethernet, Wi-Fi, or loopback.
 - **Deep Packet Inspection:** Displays detailed packet structure across multiple protocol layers.
 - **Filtering:** Supports powerful display and capture filters to isolate specific traffic.
 - **Protocol Support:** Supports hundreds of protocols (HTTP, DNS, TCP, UDP, etc.).
 - **Stream Analysis:** Allows reconstruction of TCP streams for detailed communication analysis.
- **Basic Usage Steps**
 - **Start Capture:**
Open Wireshark → Select interface → Click *Start Capturing Packets*

- **Stop Capture:**

Click the red square (Stop button)

- **Apply Display Filter:**

Use filter bar (e.g., http, dns, ip.addr == 192.168.1.1)

- **Follow Stream:**

Right-click a packet → *Follow* → TCP/UDP Stream

- **Common Filters**

- **HTTP Traffic:** http

- **DNS Queries:** dns

- **IP Address Filter:** ip.addr == <IP>

- **TCP Traffic:** tcp

- **Specific Port:** tcp.port == 80

- **Packet Structure View**

Wireshark displays packets in three panes:

- **Packet List Pane:** Summary of captured packets

- **Packet Details Pane:** Protocol breakdown (Ethernet, IP, TCP, etc.)

- **Packet Bytes Pane:** Raw data in hexadecimal and ASCII

- **Common Use Cases**

- **Network Troubleshooting:** Identify latency, packet loss, or misconfigurations

- **Security Analysis:** Detect suspicious or malicious traffic patterns

- **Protocol Analysis:** Understand how protocols behave in real environments

- **Incident Response:** Investigate breaches and analyze captured traffic

1.3.10 BURP SUITE

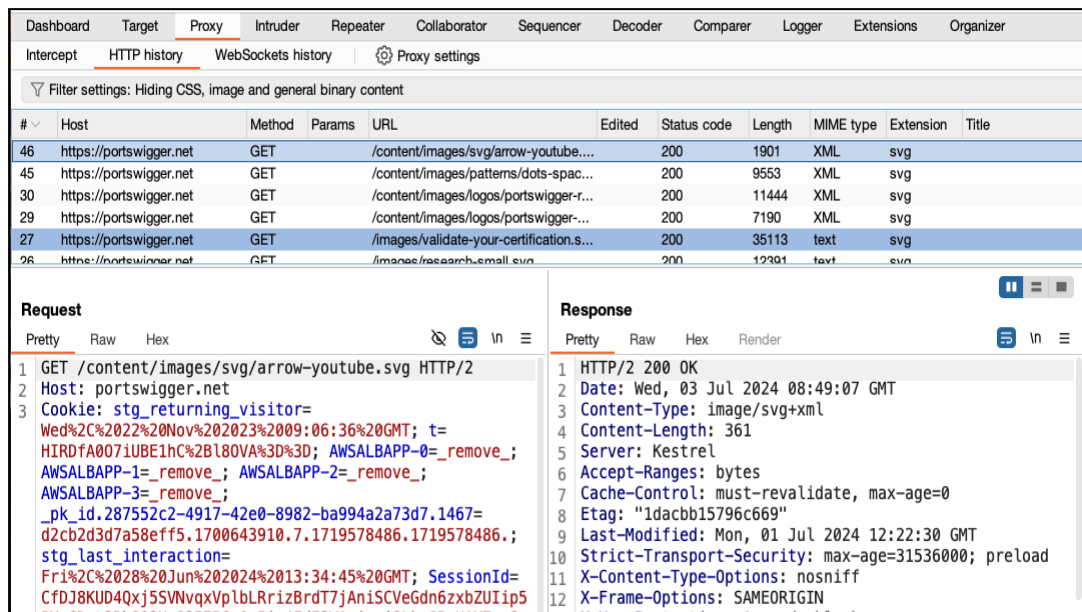


Figure 1.5 BURP SUITE

Burp Suite is a widely used web application security testing tool designed for performing penetration testing and vulnerability assessment of web applications. It acts as an intercepting proxy, allowing security professionals to analyze, modify, and replay HTTP/HTTPS requests between a browser and a web server.

Burp Suite is commonly used by penetration testers, bug bounty hunters, and SOC analysts to identify vulnerabilities such as SQL injection, XSS, authentication flaws, and misconfigurations.

The basic workflow involves configuring the browser to route traffic through Burp Proxy, capturing requests, and analyzing or modifying them.

Key Features and Components

- **Proxy:** Intercepts and modifies HTTP/HTTPS traffic between client and server.
- **Target:** Displays site map and attack surface of the web application.
- **Repeater:** Allows manual modification and resending of requests.
- **Intruder:** Performs automated attacks such as brute force and fuzzing.
- **Scanner (Pro):** Automatically detects vulnerabilities.
- **Decoder:** Encodes/decodes data (Base64, URL, etc.).
- **Comparer:** Compares request/response data for analysis.

Basic Usage Steps

- **Start Burp Suite:**

Launch Burp Suite and open a temporary or saved project

- **Configure Proxy:**

Set browser proxy to 127.0.0.1:8080

- **Intercept Traffic:**

Enable *Intercept ON* in Proxy tab

- **Forward/Modify Requests:**

Analyze and edit requests before forwarding

- **Send to Tools:**

Right-click → Send to Repeater / Intruder for further testing

Common Actions

- **Intercept Requests:** Capture login forms, API calls, etc.
- **Modify Parameters:** Change inputs to test validation
- **Replay Requests:** Use Repeater to test responses
- **Automated Attacks:** Use Intruder for payload injection
- **Analyze Responses:** Identify sensitive data or vulnerabilities

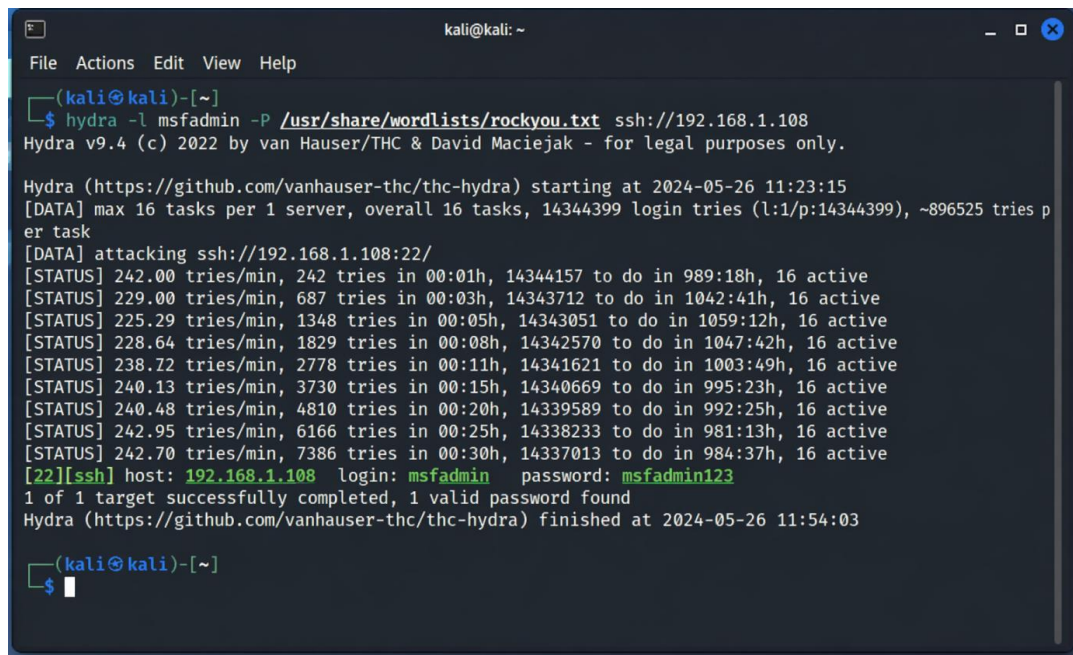
Common Use Cases

- **Web Vulnerability Testing:** Detect SQL Injection, XSS, CSRF
- **Authentication Testing:** Analyze login mechanisms and session handling
- **API Security Testing:** Inspect REST/GraphQL requests
- **Session Analysis:** Identify insecure cookies or tokens
- **Bug Bounty Hunting:** Discover real-world vulnerabilities

Example Workflow

1. Configure browser to use Burp Proxy
2. Capture a login request
3. Send the request to Repeater
4. Modify parameters (e.g., username/password)
5. Analyze server responses for vulnerabilities

1.3.11 HYDRA



```
(kali@kali)-[~]
$ hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.108
Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - for legal purposes only.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-05-26 11:23:15
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.1.108:22/
[STATUS] 242.00 tries/min, 242 tries in 00:01h, 14344157 to do in 989:18h, 16 active
[STATUS] 229.00 tries/min, 687 tries in 00:03h, 14343712 to do in 1042:41h, 16 active
[STATUS] 225.29 tries/min, 1348 tries in 00:05h, 14343051 to do in 1059:12h, 16 active
[STATUS] 228.64 tries/min, 1829 tries in 00:08h, 14342570 to do in 1047:42h, 16 active
[STATUS] 238.72 tries/min, 2778 tries in 00:11h, 14341621 to do in 1003:49h, 16 active
[STATUS] 240.13 tries/min, 3730 tries in 00:15h, 14340669 to do in 995:23h, 16 active
[STATUS] 240.48 tries/min, 4810 tries in 00:20h, 14339589 to do in 992:25h, 16 active
[STATUS] 242.95 tries/min, 6166 tries in 00:25h, 14338233 to do in 981:13h, 16 active
[STATUS] 242.70 tries/min, 7386 tries in 00:30h, 14337013 to do in 984:37h, 16 active
[22][ssh] host: 192.168.1.108 login: msfadmin password: msfadmin123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2024-05-26 11:54:03

(kali@kali)-[~]
$
```

Figure 1.6 HYDRA

Hydra is a powerful password-cracking tool used to perform brute-force attacks on various network services such as SSH, FTP, HTTP, and more. It is widely used in penetration testing to test the strength of authentication mechanisms.

In this project, Hydra is used to perform brute-force attacks on the SSH service of the target machine. It generates multiple login attempts using a wordlist, which helps simulate real-world attack scenarios. These attempts create authentication logs that are later analyzed in Splunk SIEM to detect suspicious activity.

1.3.12 CISCO PACKET TRACER

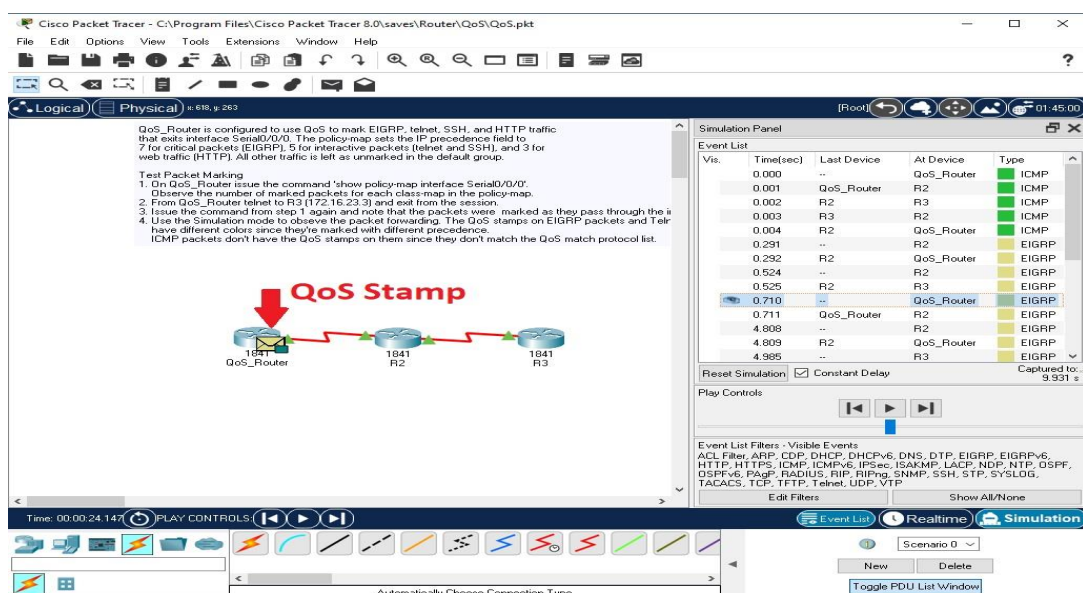


Figure 1.7 CISCO PACKET TRACER

Cisco Packet Tracer is a network simulation and visualization tool used for designing, configuring, and troubleshooting network topologies in a virtual environment. It is widely used by students and professionals to practice networking concepts without requiring physical hardware.

Packet Tracer allows users to simulate real-world networks using devices such as routers, switches, PCs, and servers, and supports configuration using **Cisco IOS commands**.

Key Features

- **Network Simulation:** Create and test network topologies virtually
- **Device Configuration:** Configure routers, switches, and end devices
- **Real-Time & Simulation Mode:** Analyze traffic flow step-by-step
- **Protocol Support:** Supports routing protocols (RIP, OSPF, EIGRP)
- **Visualization:** Graphical representation of network connections
- **IoT Support:** Simulate IoT devices and smart environments

Basic Usage Steps

- **Create Topology:**
Drag and drop devices (routers, switches, PCs) into the workspace
- **Connect Devices:**
Use appropriate cables (straight-through, crossover, etc.)
- **Configure Devices:**
Access CLI → Enter IOS commands
- **Test Connectivity:**
Use ping or simulation mode to verify communication

Common Commands (Cisco IOS)

- **Enter Configuration Mode:**
enable
configure terminal
- **Set Hostname:**
hostname Router1
- **Assign IP Address:**
interface fastethernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

- **Save Configuration:**

write memory

Common Use Cases

- **Learning Networking Concepts:** Practice subnetting, routing, switching
- **Network Design:** Build and test network architectures
- **Troubleshooting:** Identify configuration errors in a simulated setup
- **Certification Preparation:** Useful for CCNA and other Cisco exams
- **Protocol Testing:** Analyze routing behavior in different scenarios

Simulation Mode Features

- Visualizes packet flow across the network
- Shows protocol interactions step-by-step
- Helps understand ARP, DNS, TCP/IP processes

1.3.13 SIEM TOOL – SPLUNK

Security Information and Event Management (SIEM) is a cybersecurity solution used to collect, monitor, and analyze log data from multiple systems in real time, enabling organizations to detect, investigate, and respond to security threats by correlating events from different sources. It plays a crucial role in modern Security Operations Centers (SOC) by providing centralized visibility into system activities and potential security incidents. Splunk Enterprise is a widely used SIEM tool that enhances these capabilities by enabling real-time log collection, indexing, and analysis of machine-generated data from servers, applications, and network devices. It utilizes Search Processing Language (SPL) to efficiently query and analyze logs, making it easier to identify patterns, anomalies, and potential security threats.

Features of Splunk

- **Real-Time Monitoring:** Splunk continuously monitors incoming logs and provides real-time insights into system activities.
- **Log Indexing and Storage:** It efficiently stores and organizes large volumes of log data for quick retrieval and analysis.
- **Search and Analysis (SPL):** Splunk uses SPL (Search Processing Language) to filter and analyze logs based on specific conditions.
- **Visualization and Dashboards:** It provides graphical dashboards and reports to visualize security events and trends.
- **Alerting System:** Splunk can generate alerts when suspicious activities or predefined conditions are detected.

1.3.14 LABS

A cybersecurity lab is a controlled, isolated virtual or physical environment designed for testing, practicing, and researching security threats, vulnerabilities, and defenses without risking production systems. These labs enable hands-on experience in penetration testing, incident response, and malware analysis, utilizing virtual machines and simulated networks.

- **TryHackMe:** Provides structured, guided learning paths suitable for beginners to advanced users, focusing on "short, gamified real-world labs".

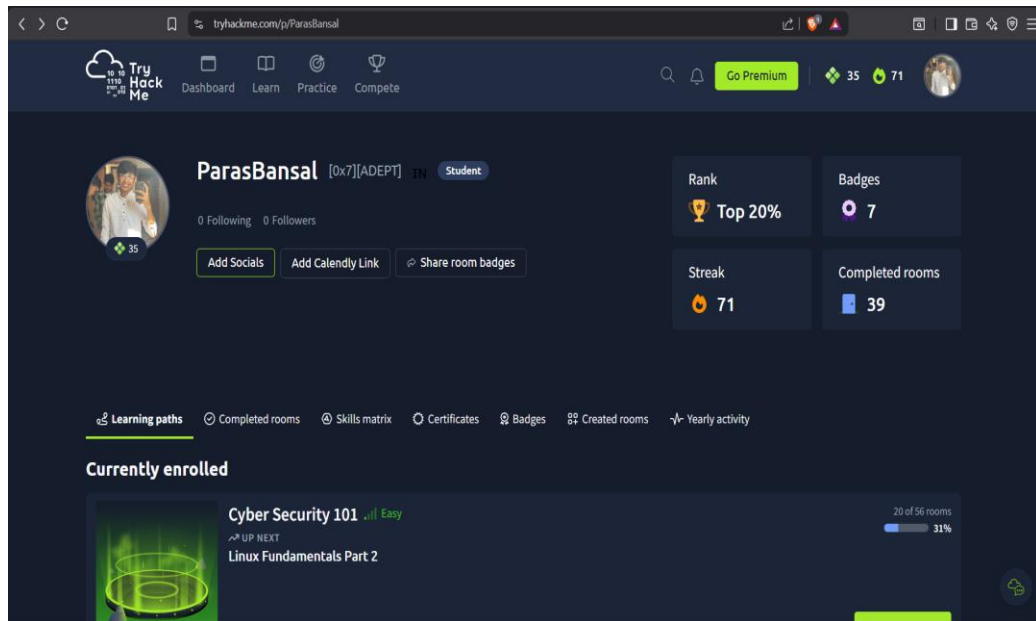


Figure 1.8 TRYHACKME

PART -2 THE PROJECT

NAME OF PROJECT

**SOC SIMULATION LAB CREATED USING VIRTUAL
MACHINES AND SIEM TOOLS**

CHAPTER-2: PROJECT DETAILS

2.1 INTRODUCTION TO PROJECT

The increasing dependence on digital technologies and networked systems has led to a significant rise in cybersecurity threats such as unauthorized access, brute-force attacks, and network intrusions. Organizations require continuous monitoring and analysis of system activities to detect and respond to such threats effectively. A Security Operations Center (SOC) plays a vital role in achieving this by providing centralized monitoring, threat detection, and incident response using advanced tools like Security Information and Event Management (SIEM) systems.

This project, titled **“SOC Simulation Lab Created Using Virtual Machines and SIEM Tools”**, focuses on designing and implementing a simulated SOC environment in a controlled virtual setup. The lab is built using VMware, where multiple virtual machines are configured to replicate real-world cybersecurity scenarios. Kali Linux is used as the attacker machine to perform various attacks, while Metasploitable 2 serves as the vulnerable target system. Splunk is used as the SIEM tool to collect, monitor, and analyze system logs generated during these activities. In this project, practical attack techniques such as network scanning using Nmap and brute-force attacks using Hydra are carried out on the target system. These activities generate logs that are collected from system directories like /var/log. The logs are then imported into Splunk, where they are indexed and analyzed using search queries and visualizations. Through this process, suspicious activities such as failed login attempts, successful logins, and potential intrusion patterns are identified. The primary objective of this project is to provide a hands-on understanding of how a SOC operates in real-world environments. By simulating both attack and defense mechanisms, the project demonstrates how cybersecurity professionals detect threats, analyze logs, and respond to incidents using SIEM tools. This lab-based approach enhances practical knowledge and prepares individuals for real-world roles in cybersecurity and SOC operations.

2.1.1 PROBLEM STATEMENT

The rapid growth of digital technologies and networked systems has significantly increased the number and complexity of cyber threats such as unauthorized access, brute-force attacks, and network intrusions. Organizations today face major challenges in continuously monitoring their systems, detecting security incidents in real time, and responding effectively to potential threats. Without proper monitoring mechanisms, these attacks can go unnoticed, leading to data breaches, system compromise, and financial loss.

A major challenge in cybersecurity education is the lack of practical exposure to real-world Security Operations Center (SOC) environments. Most learners rely on theoretical knowledge without understanding how attacks are actually performed and detected in real systems. Additionally, implementing a real SOC infrastructure is expensive and complex, making it difficult for students and beginners to gain hands-on experience.

There is a need for a cost-effective and controlled environment that can simulate real-world cyberattacks and demonstrate how security events are generated, collected, and analyzed. Such an environment should allow safe execution of attacks without affecting real systems while providing visibility into logs and system behavior.

To address these challenges, this project focuses on developing a SOC Simulation Lab using virtual machines and SIEM tools. The objective is to create a virtual environment where attacks can be simulated, logs can be generated, and security events can be analyzed using a SIEM platform like Splunk. This helps in understanding the complete workflow of a SOC, including threat detection, log analysis, and incident monitoring in a practical and secure manner.

2.1.2 OBJECTIVES OF THE PROJECT

The main objective of the SOC Simulation Lab is to design and implement a virtual Security Operations Center environment that demonstrates real-world cybersecurity operations using virtual machines and SIEM tools. The project focuses on providing practical exposure to attack simulation, log monitoring, and threat detection.

- To create a virtual lab environment using VMware for simulating real-world network scenarios.
- To configure multiple virtual machines such as Kali Linux, Ubuntu Server, and Metasploitable to perform different roles (attacker, target, and monitoring system).
- To simulate cyberattacks such as network scanning and brute-force attacks using tools like Nmap and Hydra.
- To generate system logs (authentication logs and system logs) from target machines during attack execution.

- To collect and forward logs to a centralized SIEM platform (Splunk) for monitoring and analysis.
- To analyze logs using Splunk's Search Processing Language (SPL) to detect suspicious activities such as failed login attempts and unauthorized access.
- To visualize security events using dashboards and reports for better understanding of attack patterns.
- To understand the working of a Security Operations Center (SOC) and its role in real-time threat detection and monitoring.
- To provide hands-on experience with industry-relevant cybersecurity tools and technologies.

2.1.3 PROPOSED SOLUTION

To address the challenges of limited practical exposure to real-world cybersecurity operations and the complexity of implementing a full-scale Security Operations Center (SOC), this project proposes a cost-effective and controlled SOC Simulation Lab using virtualization and SIEM technology.

The proposed solution involves creating a virtual environment using VMware Workstation, where multiple systems are deployed to simulate real-world network architecture. Different virtual machines are assigned specific roles, including Kali Linux as the attacker machine, Metasploitable as the vulnerable target system, and Ubuntu Server as the log-generating and monitoring system. This setup ensures a safe and isolated environment for performing cybersecurity experiments without affecting real systems.

In this solution, cyberattacks such as network scanning and brute-force login attempts are generated using tools like Nmap and Hydra from the Kali Linux machine. These attacks simulate real-world threat scenarios and produce logs on the target systems. The generated logs, including authentication logs and system logs, are then collected from the target machines.

To enable centralized monitoring, the logs are forwarded to the Splunk SIEM platform. Splunk is configured to receive, index, and store logs from different sources. It provides a unified interface for analyzing system activities and identifying potential security threats. Using Splunk's Search Processing Language (SPL), logs are filtered and analyzed to detect suspicious patterns such as repeated failed login attempts, unauthorized access, and abnormal system behavior.

The proposed solution also includes visualization of security events using dashboards and reports within Splunk. These visualizations help in understanding attack patterns, system behavior, and overall security status. This makes it easier to monitor and investigate incidents effectively.

Overall, the proposed solution provides a practical and scalable approach to simulate SOC operations, enabling attack simulation, log collection, analysis, and threat

detection in a single integrated environment. It serves as an effective platform for learning, experimentation, and understanding real-world cybersecurity practices.

2.1.4 MODULE DESCRIPTION

The SOC Simulation Lab is divided into four main modules, each responsible for a specific function in the overall system. These modules collectively simulate the working of a real Security Operations Center (SOC), including attack generation, log management, analysis, and monitoring.

1. Virtualization and Network Module

This module is responsible for setting up the virtual environment using VMware Workstation. It includes the creation and configuration of virtual machines such as Kali Linux, Ubuntu Server, and Metasploitable. It also handles network configuration, enabling communication between all machines. This module forms the foundation of the SOC lab by providing an isolated and controlled environment.

2. Attack Simulation Module

This module is implemented using Kali Linux and is responsible for generating cyberattacks. Tools such as Nmap are used for network scanning to identify open ports and services, while Hydra is used for performing brute-force attacks on services like SSH. These attacks simulate real-world threat scenarios and generate security events required for analysis.

3. Log Collection and SIEM Analysis Module

This module handles the collection, forwarding, and analysis of logs. Logs generated from target systems (Ubuntu/Metasploitable), such as authentication logs (auth.log) and system logs (syslog), are collected and forwarded to Splunk SIEM. Splunk indexes and analyzes these logs using SPL queries to identify suspicious activities such as failed login attempts and unauthorized access.

4. Detection and Visualization Module

This module focuses on detecting security threats and presenting the results in a user-friendly format. Based on log analysis, suspicious activities are identified and classified as potential security incidents. Splunk dashboards and visualizations are used to display logs, attack patterns, and system behavior, enabling effective monitoring and decision-making.

2.1.5 COMPONENTS OF THE PROJECT

The successful implementation of the SOC Simulation Lab requires a set of software tools and platforms that support virtualization, attack simulation, log generation, and security monitoring. These software requirements ensure proper functionality, compatibility, and performance of the entire system.

1. Operating Systems

Multiple operating systems are required to simulate different roles within the SOC environment. Kali Linux is used for penetration testing and attack simulation, Ubuntu Server is used for log generation and SIEM deployment, and Metasploitable is used as a vulnerable target system. These operating systems enable realistic attack and defense scenarios.

2. Virtualization Software

A virtualization platform such as VMware Workstation is required to create and manage multiple virtual machines on a single host system. It allows isolated environments, network configuration, and resource allocation, which are essential for a real-world SOC interface.

3. SIEM Software

Splunk Enterprise is used as the Security Information and Event Management (SIEM) tool. It is required for collecting, indexing, and analyzing logs from different sources. Splunk provides powerful search capabilities and real-time monitoring.

4. Log Collection and Forwarding Tools

Log forwarding mechanisms such as Splunk Universal Forwarder or built-in syslog services are required to send logs from source machines (Ubuntu/Metasploitable) to the SIEM system. Proper log forwarding ensures centralized monitoring and analysis.

5. Penetration Testing Tools

Security testing tools such as Nmap and Hydra (available in Kali Linux) are required to simulate attacks like port scanning and brute-force login attempts. These tools help generate realistic security events for analysis.

6. Supporting Software and Utilities

Additional software such as terminal interfaces, SSH services, and network utilities are required for system configuration, remote access, and communication between virtual machines. These tools ensure smooth operation and management of the SOC lab.

2.2 BLOCK DIAGRAM OF PROJECT

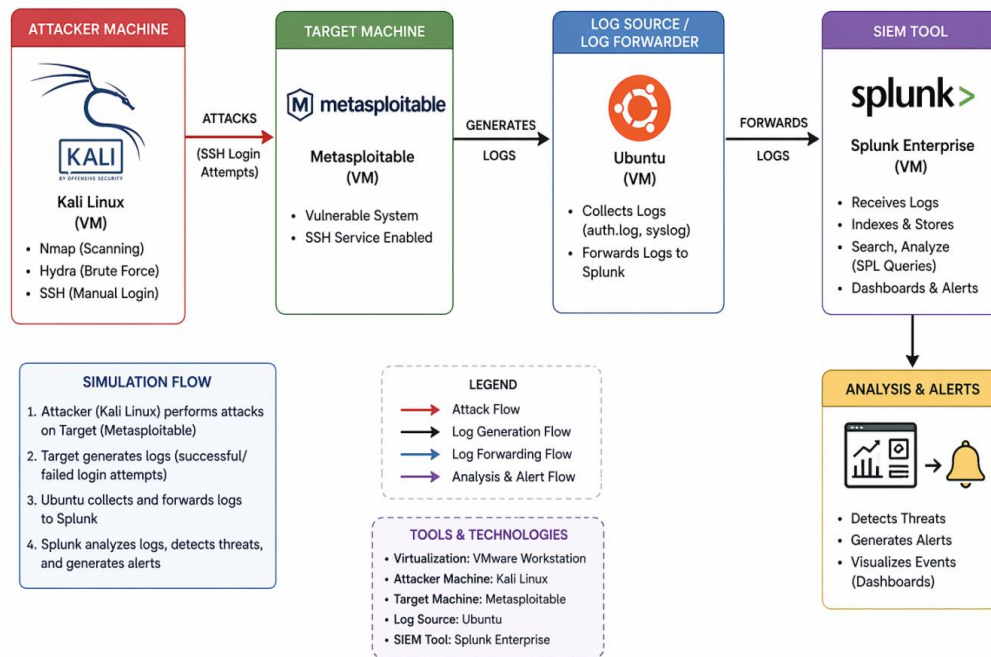


Figure 2.1 BLOCK DIAGRAM

The block diagram of the project represents the overall architecture of the SOC simulation lab created using virtual machines and SIEM tools. It illustrates how different components interact to perform network monitoring and intrusion detection.

In this setup, Kali Linux acts as the attacker machine, which performs activities such as network scanning and brute-force attacks. The Metasploitable 2 machine serves as the vulnerable target system, where these attacks are executed. The logs generated from these activities are stored in system log files such as `/var/log`.

These logs are then collected and forwarded to the Splunk SIEM system. Splunk processes, indexes, and analyzes the log data to identify suspicious activities such as failed login attempts, successful logins, and unusual patterns.

This block diagram helps in understanding the complete workflow of the project, from attack simulation to log collection, analysis, and intrusion detection within a SOC environment.

2.3 FLOWCHART OF THE PROJECT

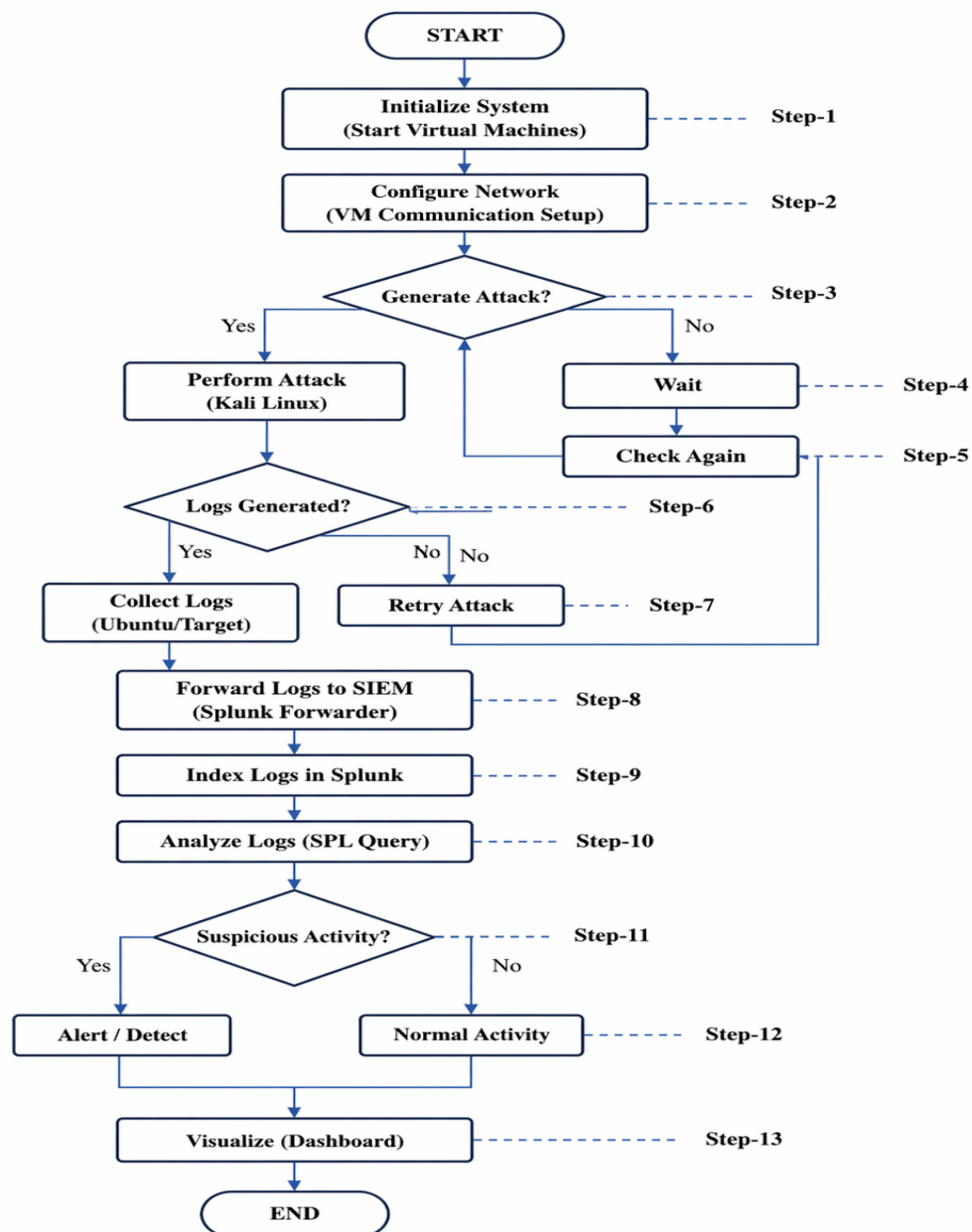


Figure 2.2 FLOW CHART OF PROJECT

The following steps describe the working of the SOC Simulation Lab:

Step-1: System Initialization

The process begins by initializing the SOC lab environment.

All required resources are prepared, including system configuration and virtual environment setup.

Step-2: Start Virtual Machines

Virtual machines such as Kali Linux, Ubuntu, and Metasploitable are started using VMware.

Each machine is assigned its role (attacker, target, and monitoring system).

Step-3: Configure Network

Network communication is established between all virtual machines.

This ensures that attack traffic can flow from Kali Linux to target systems.

Step-4: Attack Generation Check

The system checks whether an attack is initiated from Kali Linux.

If no attack is generated, the system waits and continues monitoring.

Step-5: Wait / Monitor

In case no attack is detected, the system remains in a monitoring state.

It continuously checks for any new activity or attack attempts.

Step-6: Logs Generated Check

Once an attack is performed, the system checks whether logs are generated in the target system.

Logs include authentication logs and system logs that record the attack activity.

Step-7: Retry Attack

If logs are not generated properly, the attack is repeated.

This ensures that valid logs are created for analysis.

Step-8: Log Collection and Forwarding

Generated logs are collected from the target systems (Ubuntu/Metasploitable).

These logs are forwarded to Splunk SIEM using log forwarding mechanisms.

Step-9: Log Indexing in Splunk

Splunk receives and indexes the logs into its database.

This allows efficient searching and filtering of log data.

Step-10: Log Analysis

Logs are analyzed using Splunk's Search Processing Language (SPL).

Queries are used to identify patterns such as failed login attempts or suspicious IP activity.

Step-11: Suspicious Activity Detection

The system checks whether the analyzed logs indicate any suspicious behavior.

For example, multiple failed login attempts may indicate a brute-force attack.

Step-12: Alert or Normal Activity

If suspicious activity is detected, an alert is generated indicating a potential security threat.

If no threat is detected, the activity is considered normal.

Step-13: Visualization (Dashboard)

The analyzed results are visualized using Splunk dashboards.

Graphs and logs help in understanding attack patterns and system behavior.

End

The process completes one cycle of monitoring and detection.

The system continues monitoring for future activities.

2.4 WORKING OF THE PROJECT

The SOC Simulation Lab operates by simulating real-world cyberattacks in a controlled virtual environment and analyzing the generated logs using a SIEM tool. The entire system is implemented using virtual machines created through VMware, where each machine performs a specific role within the Security Operations Center (SOC) architecture.

The working of the project begins with the initialization of the virtual environment, where multiple virtual machines such as Kali Linux, Ubuntu Server, and Metasploitable are started. Kali Linux acts as the attacker machine, while Metasploitable serves as the vulnerable target system. Ubuntu Server is used as the log-generating system and also hosts the Splunk SIEM tool for centralized monitoring and analysis.

Once the environment is set up, cyberattacks are simulated from the Kali Linux machine. Tools such as Nmap are used to perform network scanning to identify open ports and services on the target system. Additionally, Hydra is used to perform brute-force attacks on services like SSH, generating multiple login attempts. These activities simulate real-world attack scenarios and create security events within the system.

As these attacks are executed, the target systems generate logs that record detailed information about system activities. These logs include authentication logs (auth.log) and system logs (syslog), which capture events such as failed login attempts, successful logins, and user sessions. These logs serve as the primary data source for security analysis.

The generated logs are then collected and forwarded to the Splunk SIEM platform. Splunk receives and indexes the log data, organizing it into a structured format that can be efficiently searched and analyzed. This centralized log management is a key feature of SOC operations.

After indexing, the logs are analyzed using Splunk's Search Processing Language (SPL). Queries are executed to filter and identify specific events, such as failed password attempts or suspicious IP addresses. This analysis helps in detecting potential security threats and abnormal activities within the system.

When suspicious patterns are identified, such as repeated failed login attempts from a single source, they are flagged as potential security incidents. Splunk provides visualization features such as dashboards and event timelines, which help in understanding attack patterns and system behavior. These visualizations assist in decision-making and improve the efficiency of threat detection.

Overall, the project demonstrates the complete workflow of a Security Operations Center, including attack simulation, log generation, log collection, analysis, and threat

detection. It provides a practical understanding of how cybersecurity professionals monitor systems and respond to potential threats in real-world environments.

2.4.1 SOFTWARE USED IN PROJECT

The SOC Simulation Lab is implemented using a combination of industry-relevant software tools that enable virtualization, attack simulation, log generation, and security monitoring.

1. VM-WARE WORKSTATION

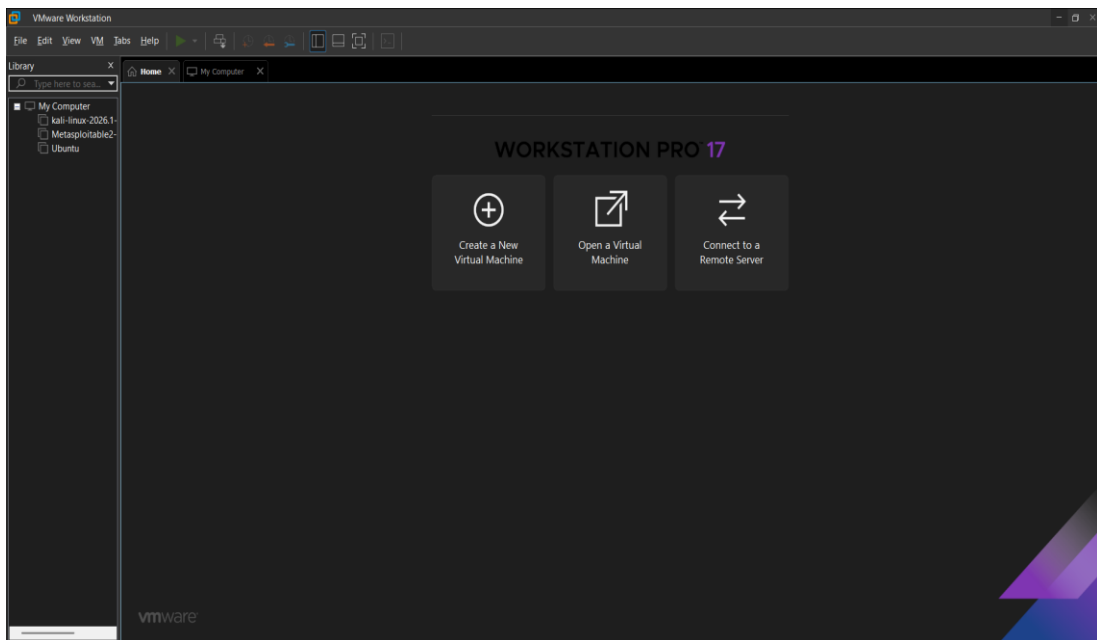


Figure 2.3 VMARE WORKSTATION

VMware Workstation is used as the core virtualization platform to design and deploy the SOC lab environment. It allows multiple virtual machines to run simultaneously on a single physical system while maintaining complete isolation between them. In this project, VMware is used to create separate environments for Kali Linux, Ubuntu, and Metasploitable. Additionally, VMware allows efficient resource allocation (CPU, RAM, storage) to each virtual machine, ensuring stable performance during attack simulation and log analysis.

2. KALI LINUX

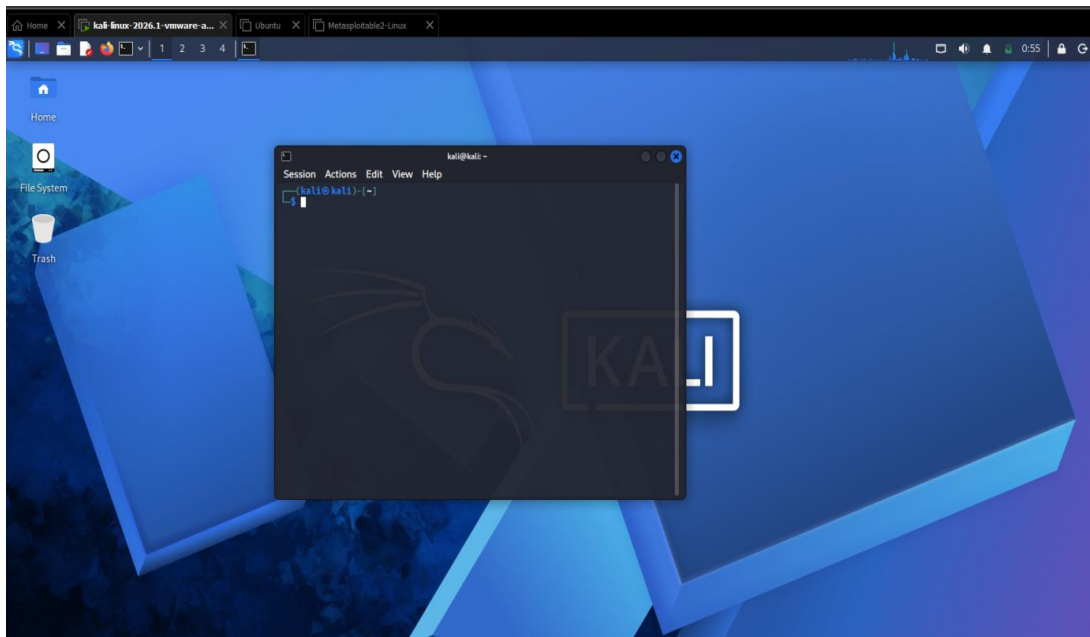


Figure 2.4 KALI LINUX

Kali Linux is used as the attacker machine in the SOC Simulation Lab. It is a specialized penetration testing operating system that comes pre-installed with a wide range of cybersecurity tools. In this project, tools such as Nmap and Hydra are used to perform network scanning and brute-force attacks.

Nmap is used to discover active hosts, open ports, and running services on the target machines. This helps in identifying potential attack points. Hydra is used to perform brute-force attacks on services like SSH, generating multiple failed login attempts. These activities simulate real-world cyberattacks.

The primary role of Kali Linux in this project is to generate attack traffic and security events. These events produce logs on the target systems, which are then captured and analyzed in Splunk. This helps in understanding how attacks are carried out and how they can be detected in a SOC environment.

3. METASPLOITABLE

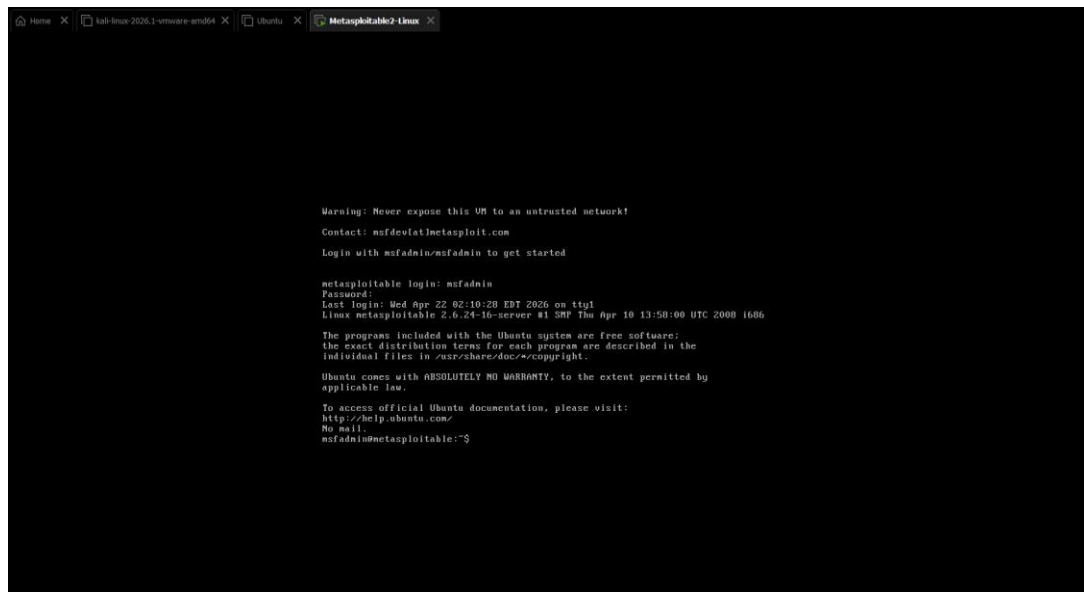


Figure 2.5 METASPLOITABLE

Metasploitable is a purposely vulnerable Linux-based virtual machine designed for security testing and training. It contains multiple known vulnerabilities, making it an ideal target for penetration testing.

In this project, Metasploitable is used as a victim system on which attacks are performed from Kali Linux. Due to its insecure configuration, it allows various types of attacks such as unauthorized access, service exploitation, and brute-force login attempts.

The main purpose of using Metasploitable is to generate realistic attack logs. These logs are critical for demonstrating how intrusion attempts appear in system logs and how they can be detected using SIEM tools. It helps simulate real-world attack scenarios in a controlled environment.

4. UBUNTU SERVER

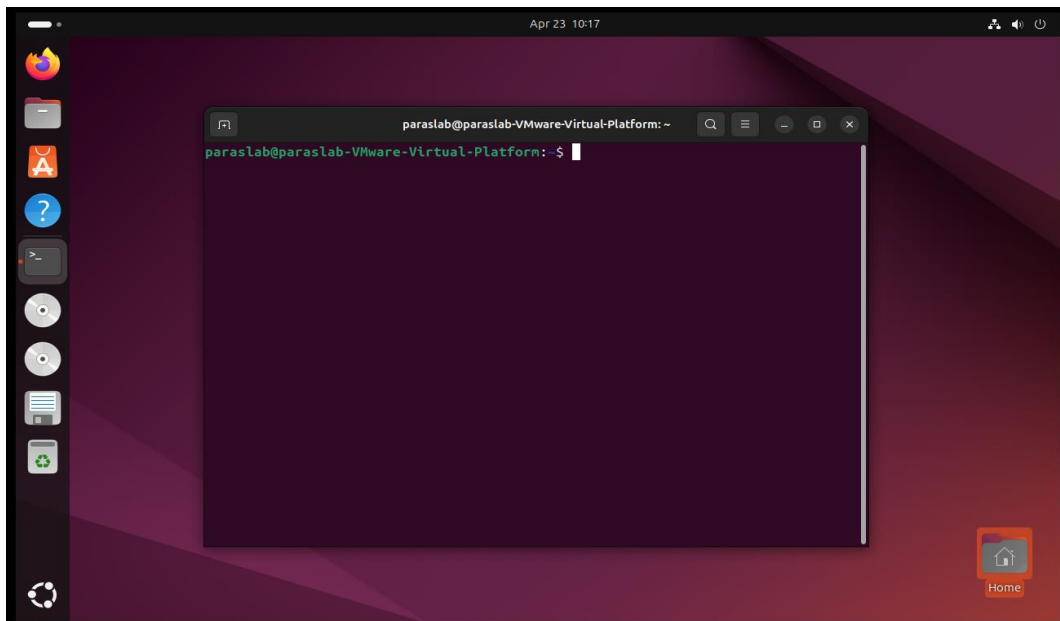


Figure 2.6 UBUNTU SERVER

Ubuntu Server is used as both a log-generating system and the host machine for the Splunk SIEM tool. It is a stable and widely used Linux distribution suitable for server-side operations.

In this project, Ubuntu generates system logs such as authentication logs (auth.log) and system activity logs (syslog). These logs record important events such as login attempts, user sessions, and system activities. When attacks are performed, these logs capture detailed information about the event.

Ubuntu is also configured to forward logs to Splunk for centralized analysis. This makes it a key component in the SOC workflow, as it acts as a bridge between raw event generation and SIEM-based analysis.

6. SPLUNK SIEM

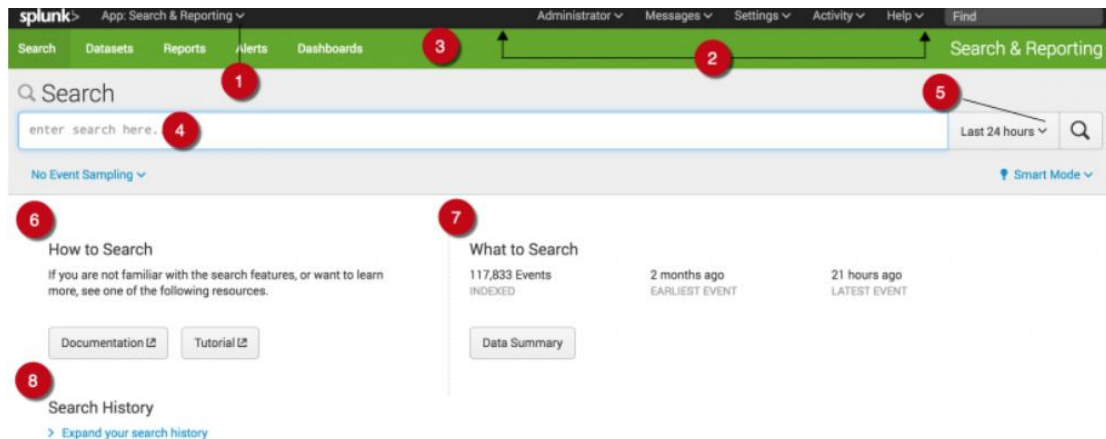


Figure 2.7 SPLUNK SIEM

Splunk is the most important component of the SOC Simulation Lab, acting as the Security Information and Event Management (SIEM) system. It is used for collecting, indexing, and analyzing logs from multiple sources in real time.

In this project, Splunk receives logs from Ubuntu and Metasploitable systems. It organizes these logs into searchable indexes and allows analysis using SPL (Search Processing Language). Queries such as detecting failed login attempts or identifying suspicious IP addresses are performed within Splunk.

Splunk also provides visualization features such as dashboards and event timelines, which help in understanding attack patterns. For example, multiple failed login attempts from a single IP can indicate a brute-force attack. This makes Splunk essential for detecting, monitoring, and analyzing security threats in the SOC lab.

2.4.2 COMMAND EXECUTION ACROSS VIRTUAL MACHINES

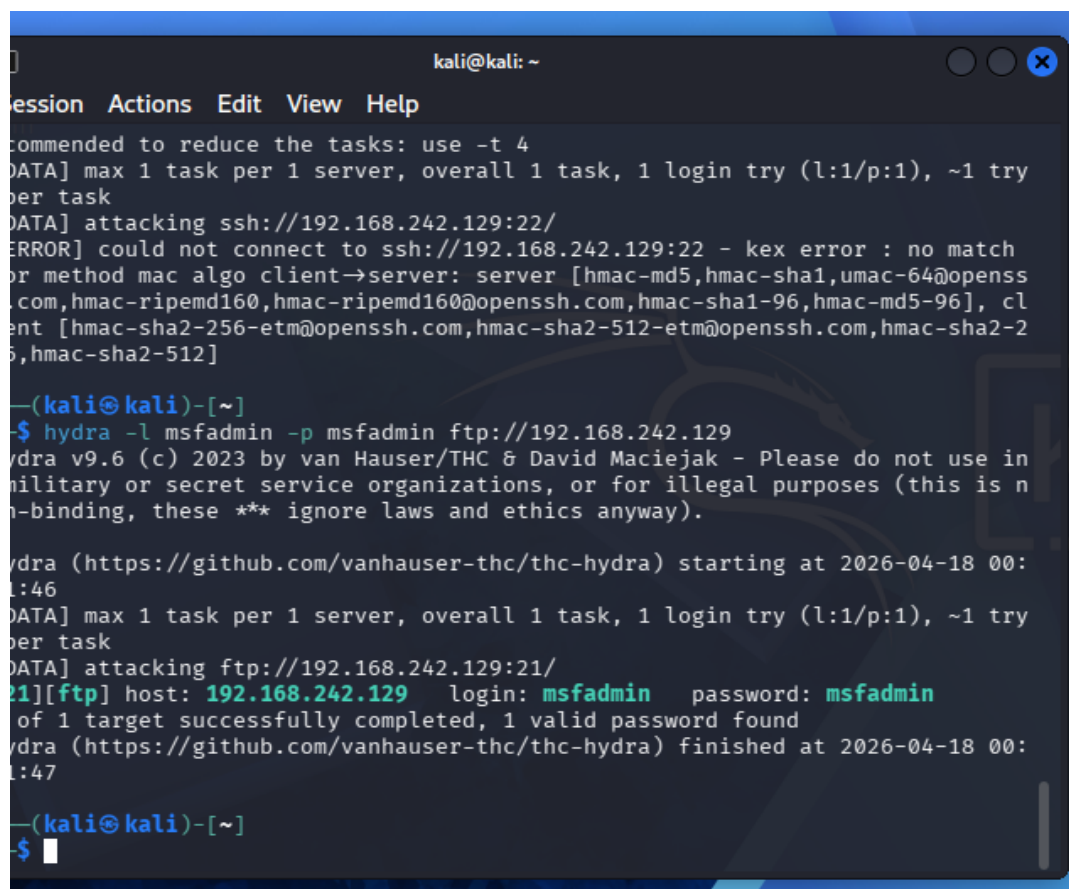
1. Kali Linux (Attacker Machine)

- `nmap -sV <192.168.1.10>`

Used to **scan open ports and identify running services** on the target machine.

- `hydra -l <username> -P <wordlist> ssh://<192.168.1.10>`

Used to **perform brute-force attack on SSH service**, generating multiple failed login attempts.



```
kali@kali: ~  
Session Actions Edit View Help  
Recommended to reduce the tasks: use -t 4  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try  
per task  
[DATA] attacking ssh://192.168.242.129:22/  
[ERROR] could not connect to ssh://192.168.242.129:22 - kex error : no match  
or method mac algo client→server: server [hmac-md5,hmac-sha1,umac-64@openss  
.com,hmac-ripemd160,hmac-ripemd160@openssh.com,hmac-sha1-96,hmac-md5-96], cl  
ient [hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha2-2  
56,hmac-sha2-512]  
  
-(kali㉿kali)-[~]  
$ hydra -l msfadmin -p msfadmin ftp://192.168.242.129  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-18 00:  
01:46  
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try  
per task  
[DATA] attacking ftp://192.168.242.129:21/  
21][ftp] host: 192.168.242.129 login: msfadmin password: msfadmin  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-18 00:  
01:47  
  
-(kali㉿kali)-[~]  
$
```

Figure 2.8 Hydra Command In Kali

2. Ubuntu Server (Log Source + SIEM Host)

- `cat /var/log/auth.log`

Used to **view authentication logs** (login attempts, failed passwords).

- `tail -f /var/log/auth.log`

Used to **monitor logs in real-time** during attack execution.

- `sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --no-prompt --seed-passwd admin123 --run-as-root`

Used to **start Splunk for the first time with initial configuration**. Automatically **accepts license, sets admin password, and runs Splunk as root** without manual input. This command is typically used during **initial installation and setup of Splunk SIEM**.

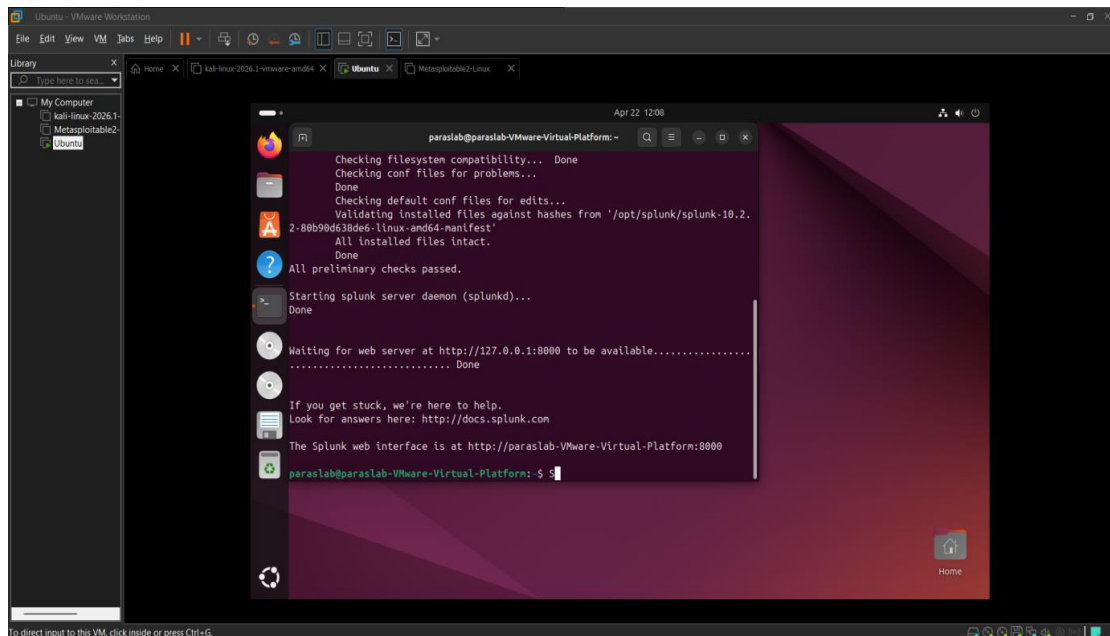


Figure 2.9 SIEM Tool Execution Command

2.5 SNAPSHOT OF THE PROJECT

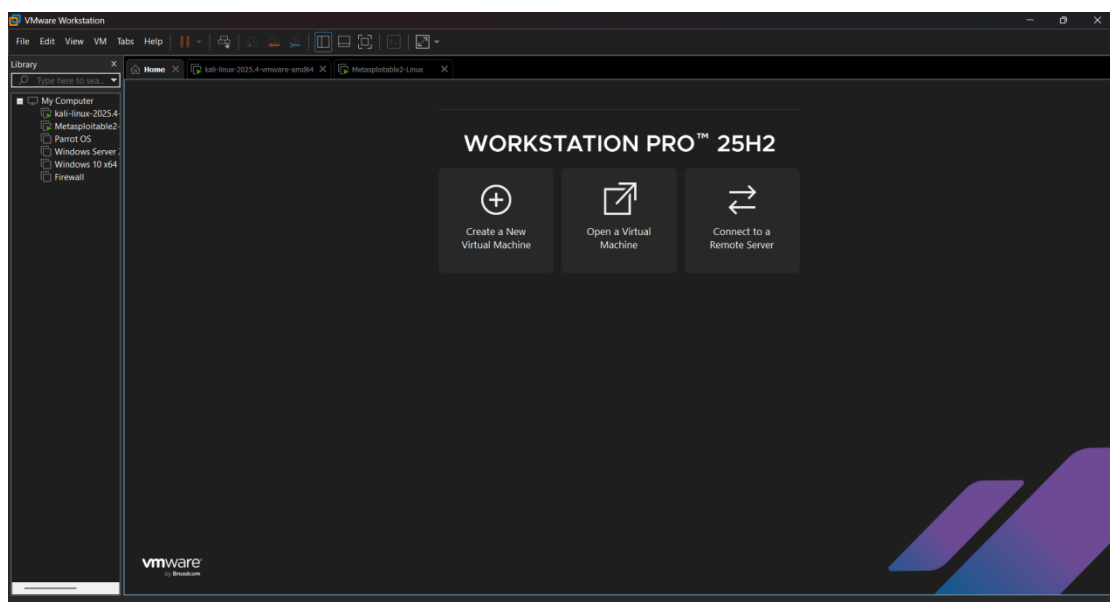


Figure 2.10 VMware Interface

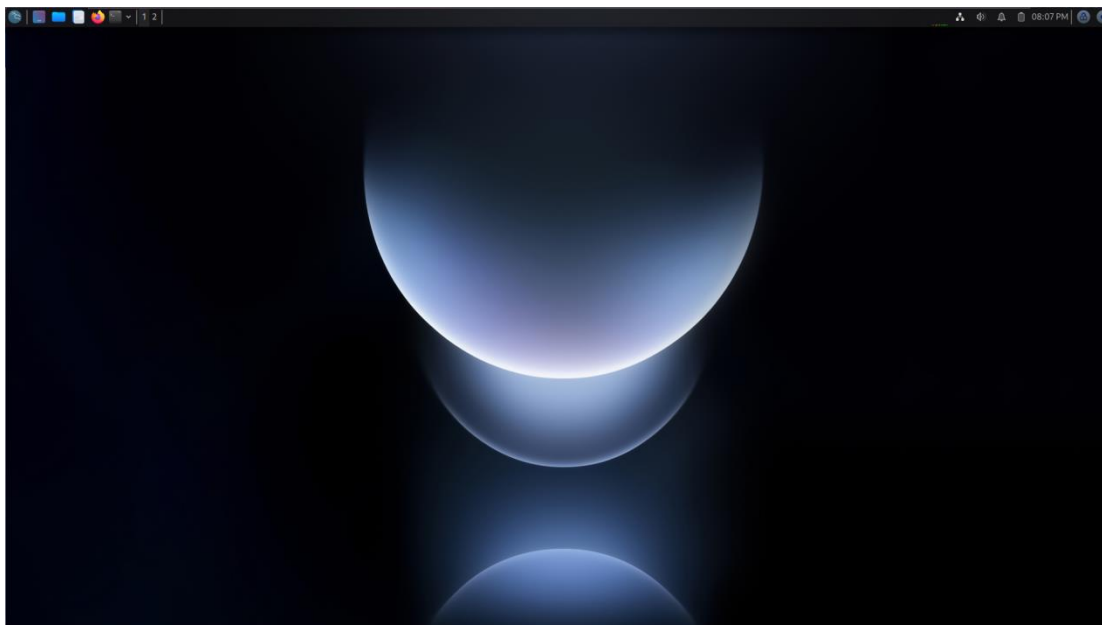


Figure 2.11 Kali Linux Interface

```
Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Sun Apr 26 06:25:52 EDT 2026 from 192.168.231.129 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Figure 2.12 Metasploitable2 Interface

```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ nmap -sV 192.168.231.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-25 23:08 +0530
Nmap scan report for 192.168.231.128
Host is up (0.0032s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login         
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:EA:C4:BF (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.10 seconds
```

Figure 2.13 Nmap scan result

2.6 APPLICATIONS OF THE PROJECT

1. Cybersecurity Training and Skill Development

The project serves as an effective learning platform for students and beginners in cybersecurity. It provides hands-on experience in performing attacks, monitoring logs, and analyzing security events. This helps in developing practical skills required for roles such as SOC Analyst, Security Analyst, and Penetration Tester.

2. Real-Time Threat Detection Practice

The lab simulates real-world cyberattacks such as brute-force login attempts and network scanning. These activities generate logs that are analyzed using Splunk SIEM. This allows users to understand how threats are detected in real time and how suspicious activities are identified in a SOC environment.

3. Security Operations Center (SOC) Simulation

This project replicates the working of a real SOC by integrating attack generation, log collection, and analysis. It demonstrates how different systems work together to monitor and respond to security incidents, making it useful for understanding enterprise-level security operations.

4. Log Monitoring and Analysis

The project highlights the importance of log monitoring in cybersecurity. Logs generated from systems such as Ubuntu and Metasploitable are analyzed to identify unauthorized access attempts and abnormal activities. This helps in understanding how logs are used for investigation and auditing.

5. Incident Detection and Investigation

Using Splunk, the project enables detection of security incidents such as multiple failed login attempts or suspicious login behavior. It also helps in understanding how incidents are investigated using log data, which is a key function of SOC teams.

6. Network Security Testing

The lab environment allows safe testing of network security by simulating attacks in a controlled setup. It helps in identifying vulnerabilities and understanding how attackers exploit systems, which is useful for improving system security.

7. Research and Experimentation

The project can be used for research purposes to study cyberattack patterns and test new detection techniques. Since the environment is virtual, different configurations and tools can be tested without affecting real systems.

8. Foundation for Advanced Security Systems

The SOC Simulation Lab can be extended by integrating advanced technologies such as cloud-based SIEM, automated response systems (SOAR), and AI-based threat detection. This makes it a strong base for future cybersecurity projects and research.

9. Incident Response Training and Simulation

The project can be used to simulate incident response scenarios in a controlled environment. By generating attacks and analyzing logs, users can practice how to respond to security incidents, investigate root causes, and take corrective actions. This helps in understanding real-world incident handling processes followed by SOC teams.

10. Centralized Log Management Implementation

The project demonstrates how logs from multiple systems can be collected and managed in a centralized SIEM platform like Splunk. This helps in understanding the importance of centralized logging in organizations, where data from different sources is aggregated for better visibility, correlation, and security monitoring.

2.7 REFERENCES

1. Splunk Enterprise Documentation,
Available at: <https://docs.splunk.com>
2. Kali Linux Official Website,
Available at: <https://www.kali.org>
3. VMware Workstation Documentation,
Available at: <https://www.vmware.com>
4. Metasploitable Documentation,
Available at: <https://sourceforge.net/projects/metasploitable>
5. Ubuntu Server Documentation,
Available at: <https://ubuntu.com>
6. Nmap Reference Guide,
Available at: <https://nmap.org>
7. THC Hydra Documentation,
Available at: <https://github.com/vanhauser-thc/thc-hydra>
8. Cybersecurity Learning Resources and Tutorials,
Various online sources and practical lab references.